BrickPlan: A Benchmark for Visual Assembly and Error Recovery Supplementary Material

Anonymous CVPR submission

Paper ID Draft

A. Artifact Scope and Reproducibility

The manuscript describes an evidence snapshot dated 21 October 2026. All earlier models are included through their current versions. Three banks coexist: the 3,292-question white-baseplate static bank, the 1,872-episode dynamic mechanism bank, and the 96-case community module suite. Their counts are not independent model populations, and their experimental results are not interchangeable. The new route study runs on the community module suite, not on the historical six-part car. The complete module suite has no formal learned-agent evaluation or independent human assembly experiment in this snapshot.

The artifact records original model files, source metadata, certificates, cases, reference manuals, intervention witnesses, and order-regression checks. Each instruction contains the physical part and the resulting scene. Image manifests bind the current benchmark renders to their source states. Paper figure provenance separately records source hashes, cameras, native render dimensions, and exported figure hashes. The deterministic study records the four input files for each object and their SHA-256 digests. Re-execution does not require a model API.

The public object gallery is an author-facing inspection tool: it shows fault names and correction witnesses. It is not a valid blind agent input. Evaluation must use the observation adapter, which exposes the permitted images and action handles while retaining certificates, intervention labels, event timing, and recovery witnesses privately. Publishing such annotations supports reproducibility but also makes these public examples unsuitable as an undisclosed held-out test set.

B. Complete Model Coverage

The three tracks contain 5,260 current records: 3,340 static and 1,920 dynamic. Their 240 configuration entries are not independent objects. All 127 earlier static configurations and all 30 earlier dynamic structures remain in the current banks. The original chair, table, car, and crane are included in Figure S1; another 38 complete object designs and 26 structural configurations are also shown. Twelve community modules extend this coverage rather than replacing it. The artifact's full catalogue lists every model identifier, source, target-part count, case count, and error distribution. Tables at the end of this supplement enumerate all 160 static configurations.

C. Exact Observation and Action Roles

Table S2 distinguishes the images that define a task from those that show its current state. In a static question, the acceptance target is an intermediate stage. A full-completion reference is provided for context and must not silently replace that stage target. In a dynamic episode, the acceptance target is the completed module and remains fixed even when a hidden intervention occurs.

Views and handles. The module interface provides isometric, rear, and top views at 900×660 pixels. Part and candidate handles are randomized by seed and do not expose target labels. The manual stays accessible throughout the episode. Current-state images do not include the paper's error boxes or corrective arrows. Explicit request and response logs distinguish inspection from manipulation and retain the actual view delivered to the agent.

One physical action per turn. An attach action chooses one rack piece and one catalog candidate. A detach action removes one installed piece. The executor checks the resulting configuration and a straight insertion or extraction path. An invalid physical attempt consumes a manipulation attempt, reports a legality code, and leaves the world unchanged. Inspection does not advance construction or trigger an event. Finish requests termination; success additionally requires the exact permitted geometric endpoint. The manipulation budget is $4N + 12$ for N normal module pieces. The outer model runner has a separately configured decision cap so repeated inspection cannot run indefinitely. Both limits must be reported in a model comparison.

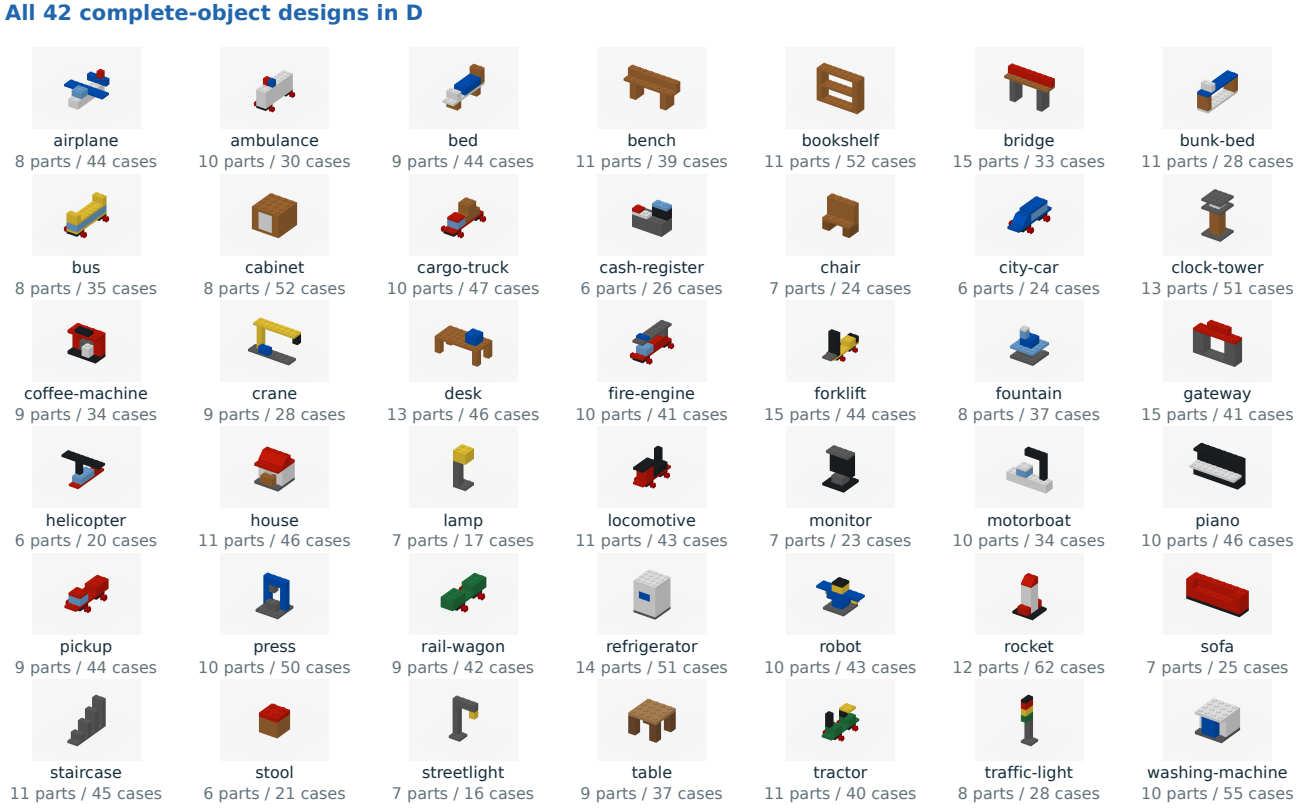

Figure S1. **All 42 complete-object designs in D.** Every object is shown with its target-part count and current episode count. These are deliberately simplified complete constructions, including the original four objects. Their geometric simplicity is not evidence of measured human or model difficulty.

Figure S2. **All 26 structural configurations in D.** These configurations provide source-derived assemblies and controlled tests of access, support, buried errors, and reconstruction. Together with Figure S1, they cover all 68 dynamic structures and 1,872 episodes.

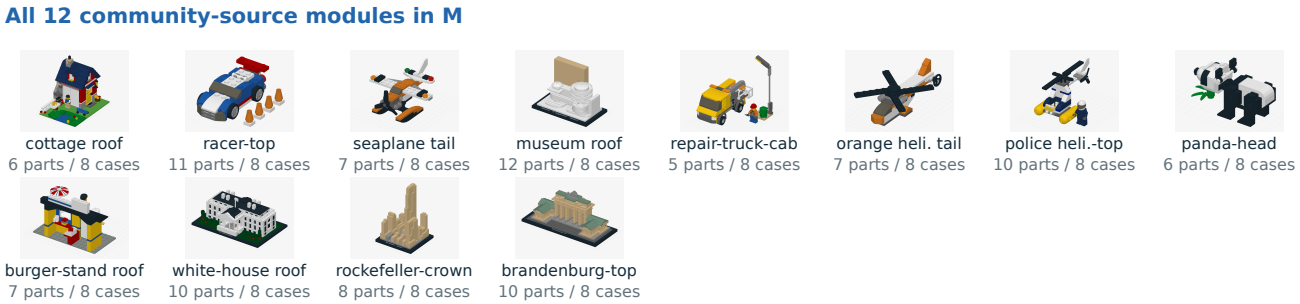

Figure S3. **All twelve community sources in M.** The entire source scene provides context; only the bounded 5–12-part module is manipulated. Part counts below the scenes count those task pieces, not the complete source object. Each module supplies eight cases.

Table S1. **Related evaluation contracts.** Inputs and outputs determine what a failure can mean. This is a task comparison, not a performance comparison.

Work	Goal information	Output / interaction	Recovery setting
BrickNet; BrickBench	Generation conditions or text	Assembly program / agentic design	Validity during generation; no prescribed external fault in a supplied module.
MEPNet; SCANet	Visual assembly manual	Part poses / step correction	Manual execution or correction of a particular assembly step.
Break and Make	Interactive inspection of a target	Inspect, disassemble, reconstruct	Reconstruction after a dedicated inspection phase.
AsgardBench; RoboRecover	Task and visual scene	Interactive actions / robot policy	Plan adaptation or continuation from execution deviations.
BRICKPLAN module suite	Prebuilt context, goal, full remaining manual	One attach, detach, inspect, or finish	Continuous construction with an actual-state intervention; any legal order is accepted.

Finite action candidates. The action catalog is part of the task definition. This is not unrestricted six-degree-of-freedom pose prediction. Candidate previews and legality codes reduce the planning burden and can leak useful physical information, even though neither states the correction. They must be held constant across compared agents; an ablation can quantify their contribution. A symbolic state or expected next-state overlay is privileged information and should be reported as a separate condition.

Indistinguishable parts. Terminal comparison uses physical part geometry, color, and pose rather than instance identity. Supported rotational equivalences are limited to plain rectangular parts with the appropriate geometric symmetry. Decorated, directional, and asymmetric pieces are not assumed invariant to arbitrary rotations. Identical-part exchanges are tested separately from sequence exchanges: changing an identifier alone must not create an apparent alternative physical trajectory.

D. Intervention Selection and Causal Comparisons

Deterministic adaptive selection. Each community fault condition names one of three mechanisms and an installed-part threshold. After an accepted manipulation reaches the threshold, the selector inspects the actual state. For a missing part or color replacement, it considers older installed normal pieces, retaining the newest piece to avoid merely undoing the last action. For an extra layer, it searches compatible exposed positions for the spare plate. The control and all three fault conditions share the same inventory and candidate catalog.

An event is admitted only if it changes the state through legal primitives and has a feasible inverse returning to the pre-event state. A replacement uses a same-shape alternative-color spare: the environment removes the original piece and attaches the spare between observations. These two environment primitives do not allow the agent to submit two manipulations in one turn. If no suitable event is available, selection is deferred. At most one event is applied in the current module suite. The private log records the realized location and its inverse.

Table S2. **Image roles in the community interface.** A reference image is not a report of current execution. These roles are explicit in the adapter rather than inferred from file order.

Role	Content	Interpretation
Scene overview	Full original object, initial and completed views	Establishes the recognizable object and the supplied context.
Module start	Cropped scene before any task manipulation	The initial movable-part state; the context is already assembled.
Full-completion reference	Completed remaining module	Explains the whole module; may differ from a static question's acceptance target.
Acceptance target	Required final state for this episode	Stage target for static repair; completed module for dynamic completion.
Instruction part / after	Isolated part and scene after its reference placement	One executable demonstration; the numbered order is not compulsory.
Inventory card	Shape and color of an available piece	Identifies an action handle without declaring its correct placement.
Before last action	Actual scene before the agent's last action	Temporal evidence; explicitly absent at the first decision.
Current scene	Actual state after execution and any intervention	The authoritative observation of the world now.
Candidate preview	Hypothetical placement of a selected piece	A planning aid; not executed, not a correctness certificate.

What feasibility establishes. A legal inverse is a recovery witness, not a proof that it is the shortest recovery or that the error is visually obvious. Every selected normal route in the protocol study has a complete replay, so restoring its pre-event state permits the remaining route to finish. This guarantee does not extend to an arbitrary agent state that was already incorrect before the intervention. Likewise, a reference event's RGB check does not certify visibility at every location an adaptive selector can reach. New trajectories require their own observation-level audit.

Matched-prefix controls. For a causal recovery comparison, execute a normal prefix once, save the environment and conversation, and branch it into control and fault continuations. The two branches have identical history before the event. Keep the target, action vocabulary, inventory, renderer, budgets, and model settings fixed. Report both completion and whether the fault actually occurred. A run that finishes correctly without encountering its scheduled fault is a completion, not a successful repair trial.

A conditional recovery rate excludes untriggered episodes from its denominator, so it must be accompanied by exposure coverage and the number of stalled prefixes. Even with the same input seed, two fresh API runs are not necessarily matched-prefix controls: their actions may diverge before the event. Adaptive location selection also creates trajectory-dependent fault difficulty. Stratify by the realized edit and report the reference-location subset separately when appropriate.

E. Community Objects and Route Diagnostics

Table S3 lists all twelve objects. Original instance counts include the supplied surrounding scene, figures, and accessories. The module count measures the actual task; it does not imply reconstructing the entire source object. The full scenes contain 2,135 source instances; the bounded modules contain 99 normal placements. Larger source models therefore need not be harder tasks.

Selection of the alternative routes. Each reference route is the verified manual. An alternative is obtained by testing swaps and retaining a physically legal route with all three feasible adaptive faults. When available, the search favors a route whose geometry at the threshold differs from the reference checkpoint. The study therefore uses selected regression examples, not a random sample of all topological orders or an estimate of agent route choice. Table S4 provides the precise reference-position indices.

Exact-prefix opportunity. The comparison inspects the otherwise normal physical trajectory for equality with the reference checkpoint. This avoids confounding trigger eligibility with a different repair policy. It yields 36/36 eligible reference-route fault trials and 9/36 alternative-route trials. The actual-state selector applies an event in all 36 trials in each group. A separate comparator requiring every intermediate state to match the manual accepts 12/12 reference routes and 0/12 alternatives. The terminal comparator accepts all 24 normal routes. These numbers establish counterexamples to an overly restrictive contract, not a learned-

Table S3. **Complete community suite.** Every row has four static and four dynamic cases: control, missing piece, wrong color, and extra layer. The threshold is a count of installed normal pieces, not a compulsory reference prefix.

Source set	Object	Remaining module	Source parts	Task parts	Threshold
31009	Cottage	Roof section	270	6	3
31027	Blue racer	Upper cabin	67	11	3
31028	Seaplane	Tail assembly	53	7	3
21004	Guggenheim Museum	Roof section	208	12	4
3179	Repair truck	Cabin top and lights	128	5	4
30181	Orange helicopter	Tail assembly	52	7	3
30311	Police helicopter	Rotor assembly	59	10	4
30026	Panda	Head section	61	6	4
6683	Burger stand	Roof and sign	79	7	3
21006	White House	Chimneys and roof details	556	10	4
21007	Rockefeller Center	Upper crown	240	8	7
21011	Brandenburg Gate	Top crest	362	10	6
Total	12 objects	96 cases	2,135	99	—

Table S4. **Alternative legal routes used in the deterministic study.** Reference-position numbers identify pieces; the actual executor uses randomized handles. All routes reach the same normal target. The threshold states differ for nine objects.

Object	Alternative route in reference-position indices	Different threshold state
Cottage	1, 2, 4, 3, 5, 6	Yes
Racer	1, 6, 3, 4, 5, 2, 7, 8, 9, 10, 11	Yes
Seaplane	1, 6, 3, 4, 5, 2, 7	Yes
Museum	1, 2, 3, 5, 4, 6, 7, 8, 9, 10, 11, 12	Yes
Repair truck	1, 2, 3, 5, 4	Yes
Orange helicopter	1, 3, 2, 4, 5, 6, 7	No
Police helicopter	1, 2, 3, 5, 4, 6, 7, 8, 9, 10	Yes
Panda	1, 2, 3, 5, 4, 6	Yes
Burger stand	2, 1, 3, 4, 5, 6, 7	No
White House	1, 2, 3, 5, 4, 6, 7, 8, 9, 10	Yes
Rockefeller Center	1, 2, 3, 5, 4, 6, 7, 8	No
Brandenburg Gate	1, 2, 3, 4, 5, 7, 6, 8, 9, 10	Yes

agent performance improvement.

The blind policy follows its normal route after any change. It completes 24/24 controls and 0/72 fault trials. The privileged policy uses the recorded inverse immediately after the event and then continues; it completes 96/96 trials. No model API is involved. A privileged repair trace tests solvability and logging, not visual detection, reasoning, or human-level assembly skill.

F. Static Migration and Error Coverage

The static bank uses an explicit white baseplate and disallows unsupported floating states or passage through the board. Its migration ledger records the five routes in Table S5. All resulting repair witnesses replay under the new executor. Figure S4 shows actual current and target states for eight error categories.

Of the 267 replacements, 250 source targets were not buildable under the new rules, 13 had board-penetrating initial states, and four were duplicate states. Replacements preserve their source links and substitution reasons. The ledger's 3,025 preserved cases include ten repaired initial scenes; "preserved" therefore does not mean that every initial coordinate was unchanged. The 160 controls have a zero-action repair witness.

Table S5. **White-baseplate migration ledger.** Counts refer to questions, not independent objects.

Migration method	Questions
Direct replay	2,144
Verified dismantle/rebuild	834
Witness using held components	37
Corrected board-penetrating initial scene	10
Replaced incompatible source case	267
Total certified questions	3,292

Table S6. **Dynamic mechanism counts.** Each episode belongs to one recorded family, including the explicitly compound family.

Condition	Episodes
Normal control	160
Removed piece	214
Shifted piece	246
Cascaded removal	160
Extra layer	179
Early cover	31
Wrong color	218
Wrong size	80
Rotated piece	132
Swapped positions	23
Wrong part type	98
Buried color error	209
Compound fault	122
Total	1,872

A complete geometric replay is a necessary check, but does not establish that every question has sufficient visual evidence or comprehensible wording. The full static RGB package and per-question human readability review are still pending. It would be incorrect to interpret the 3,292/3,292 replay count as a human-validated visual dataset size.

G. Dynamic Mechanisms and Visibility

The mechanism bank contains 68 structures: 42 simplified recognizable objects and 26 smaller structural configurations. Of its 1,872 episodes, 1,588 use complete-object models and 284 use the remaining configurations. There are 160 controls and 1,712 interventions across twelve families (Table S6). This is broader mechanism coverage than the three error families in the community suite, but does not substitute for testing those mechanisms on the larger source-preserving modules.

Machine visibility screens. Pixel differences use a maximum RGB-channel change of at least 24. The reference mechanism audit retains 1,553 interventions as primary machine candidates with at least 2,000 contrast pixels in an available view. Another 136 interventions are classified as action cancellation, and 23 have low contrast; there are no duplicate event states. A visible change in pixel space is not proof that a person can identify the incorrect part or infer a valid correction. Re-rendering and rechecking the realized event is necessary when a different route changes its location or occlusion.

The community reference snapshots use a separate minimum of 128 changed pixels in at least one fixed module view. All 36 reference fault snapshots pass that machine screen. Its lower threshold is an engineering admission criterion, not an experimentally calibrated human perceptual threshold. The two visual admission rules must not be pooled as one validated population.

Splits and difficulty. The mechanism bank groups sources into 43 families for development and evaluation separation. Its balanced development subset has 108 faults (nine per family) and 42 controls; the evaluation subset has 168 faults (fourteen per family) and 72 controls, where "family" in these counts means an error mechanism. Source-family separation applies within this bank, not across every artifact in the repository. The community suite is a public development collection and has no undisclosed held-out source-object evaluation in the current report.

Part count, prerequisite depth, access obstruction, and the amount of reconstruction needed after a fault describe different dimensions of difficulty. Neither source-file size nor the labels "roof" and "car" justify an easy/medium/hard ranking without performance evidence. Figure S6 reports case coverage without making that inference.

H. Historical Development Evidence

Earlier API car pilot. A prior interface version records three completed episodes with the provider route named `google/gemini-4-argon` on a six-part car (Table S7). These were development traces with seed 41 and a 40-decision cap. They used a different bank and interface, without the community manuals, and cannot populate a result table for the current suite. One control response was invalidly formatted; no illegal physical action was executed. A normal-run trajectory diverged from the fault runs before the intervention, so these traces are not a paired causal estimate of repair difficulty.

How to read the GT/model comparison. The main paper's eight-column filmstrip uses the extra-layer and control traces from this archive. GT is a feasible reference, not a required intermediate trajectory. The extra-layer run completes in fourteen decisions with seven accepted physical moves; its first two attachments and final two additions have a different order from GT. In the control run, P001 is legally placed at C006 rather than its target pose, then detached and reattached correctly. Physical legality therefore does not imply target correctness. All displayed episodes eventually succeed; the red cell marks a transient wrong placement, not final failure.

The figure export checks every saved state hash and feedback code in both runs (32 decisions total), and supplies a response-free JSON with the original certificate, selected states, source hashes, and actions. The plotted states use one shared camera and framing. Inspections are omitted; the final control image also follows an unshown last place-

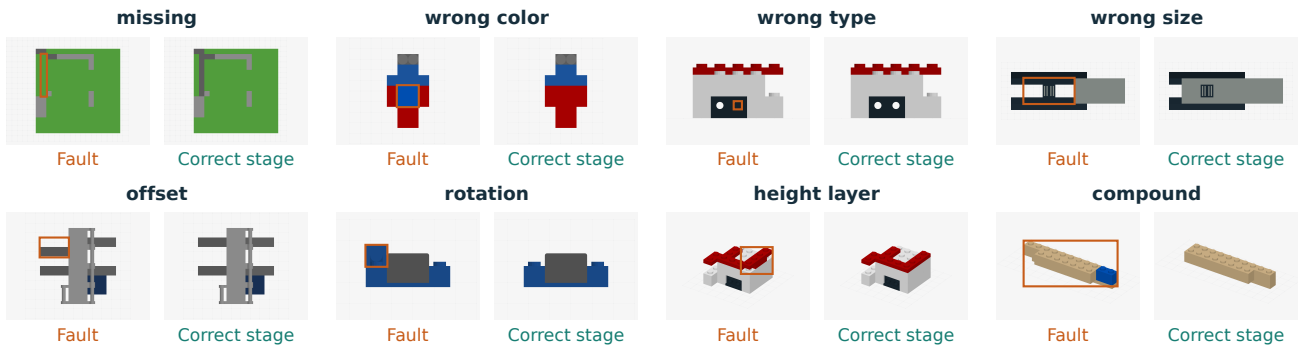

Figure S4. **Eight static error categories on the white baseplate.** Each compact cell places the faulty current state on the left and the correct stage target on the right. Categories cover missing pieces, color/type/size substitutions, translation, rotation, layer height, and compound errors. Orange difference boxes are paper annotations only.

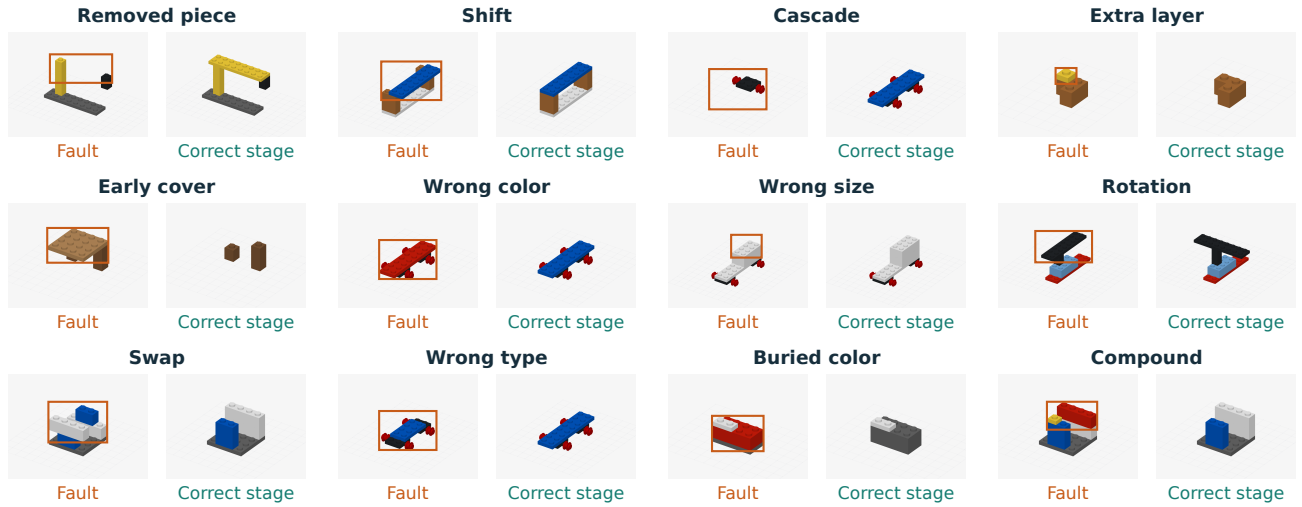

Figure S5. **All twelve dynamic error mechanisms.** Each cell compares a faulty state with its correct intermediate stage, using a common view. Examples are drawn from complete objects in D. They illustrate detectable scene changes and witnessed repairs; they do not establish human error frequencies or model detection performance. Boxes are explanatory annotations only.

Table S7. **Archived pilot only.** Costs are the recorded charges for these individual calls, not current prices or a projected experiment budget. All three completed the older car task.

Condition	Decisions	Legal moves	USD
Control	18	8	0.1950043
Extra layer	14	7	0.2022685
Removed piece	14	7	0.2140114

ment. The external extra layer is labeled separately from the model's mistake. These examples illustrate reporting conventions and cannot establish current-bank success rates or an independent effect of interventions.

Earlier no-progress runs and HTTP 403 provider failures occurred during interface development. An API access failure is not evidence of a model's assembly ability. There are zero paid model episodes in the new community suite and mechanism-v6 release record.

Earlier human check. One recorded participant answered binary change-detection questions correctly on eleven selected, unprompted images; a twelfth item had already exposed its answer and is excluded. Two further reviewers were reported to agree, but have no independent submission records. This is not a three-rater study, a timed assembly baseline, or validation of all current cases. It supports only a narrow development observation about those eleven images.

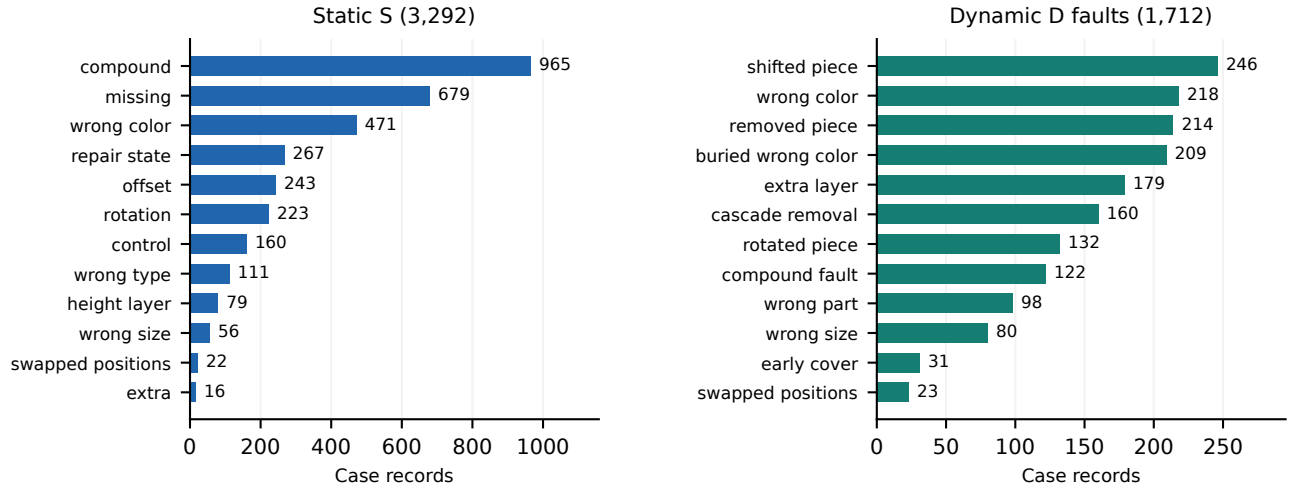

Figure S6. **Coverage of the broader banks.** Static counts include normal controls; dynamic bars show intervention families and exclude 160 controls. Counts describe repeated tasks on shared models, not independent object diversity or measured difficulty.

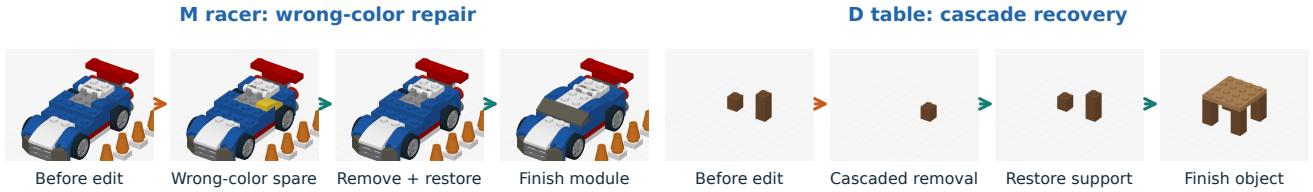

Figure S7. **Two certified repair examples, eight views in one row.** Left four: replace a wrong-color racer cabin piece and finish the community module. Right four: restore pieces removed by a cascade in the original table object. Each transition summarizes selected stages; any multi-part recovery executes separate one-brick actions. These are geometry witnesses, not model rollouts.

I. Planned Experiments and Reserved Result Tables

All dashes in the fourteen planned result tables (E1–E14) are uncollected measurements. They are neither zero scores nor executor results. The artifact contains machine-readable case assignments, participant schedules, and a results template whose empty values are `null`. These define the planned experiment; they are not an implementation claim for the paired API runner or human study.

Models and primary samples. Table S8 names ten candidates: four moderate-cost commercial APIs and six open visual checkpoints. Earlier Qwen2.5-VL and Gemma 3 models complement the newer candidates; costly flagship APIs are omitted. Official provider catalogs and model cards document image inputs; account access is not tested. Open-weight sizes support within-family comparisons, not a claim that size alone explains performance. The machine-readable roster links every row to its official source. Before execution, record the returned API revision or checkpoint commit, engine, precision, image processing, and native reasoning/decoding settings. Do not silently replace an unavailable candidate; report it as not run. Use one run per case and seed 41 for handle randomization; record a provider seed only when supported. Source-clustered intervals quantify variation across sources, not decoding stability across repeated runs. Generation/size comparisons also differ in training and architecture, so neither effect is isolated. No frontier-model claim is made. The S test list contains 467 question IDs from 20 source groups. Migration changed the source of some questions; 151 stale assignments are repaired while retaining the existing source-group allocation. The corrected current split contains 2,286 train, 539 development, and 467 test questions. D uses its existing 168-fault/72-control evaluation definitions, with fourteen faults per mechanism. M uses all 48 static cases, 36 dynamic fault definitions, and twelve dynamic controls; it remains a public development panel. The full-bank secondary report covers every model configuration and retains separate track/source averages. No joint tuning may use a source later presented as unseen in another track.

Table S8. **Named evaluation candidates.** Official catalogs/model cards checked on 23 October 2026. These are requested IDs, not verified account access or pinned served revisions. Record returned API version or checkpoint commit, inference engine, precision, image processing, and decoding settings before execution. All rows remain unmeasured.

Model	Access	Requested API ID / checkpoint
GPT-6.1 Sol	API	`gpt-6.1-sol`
Claude Sonnet 5.5	API	`claude-sonnet-5-5`
Gemini 3.8 Flash	API	`gemini-3.8-flash`
Claude Haiku 5.5	API	`claude-haiku-5-5`
Qwen3.5-9B	Open	`Qwen/Qwen3.5-9B`
Qwen3.5-27B	Open	`Qwen/Qwen3.5-27B`
InternVL3.5-8B	Open	`OpenGVLab/InternVL3_5-8B-HF`
Qwen2.5-VL-7B	Open	`Qwen/Qwen2.5-VL-7B-Instruct`
Qwen2.5-VL-32B	Open	`Qwen/Qwen2.5-VL-32B-Instruct`
Gemma 3-12B	Open	`google/gemma-3-12b-it`

Table S9. **Related experimental designs and our adaptations.** These works motivate controls and analyses; their original tasks, agents, and scores are not directly comparable to BrickPlan. E1–E14 identify our planned result tables.

Work	Experimental element	BrickPlan adaptation
LEGO-Puzzles [4]	Task-wise model and human comparison	Named candidates, common human panel, difficulty (E1/E5/E6)
PhyBlock [3]	Complexity levels and part precision/recall	Complexity and secondary part F1 (E6/E10)
AsgardBench [5]	Observation, history, and feedback ablations	Input controls and failure ledger (E3/E7/E13)
SCANet [6]	Correction versus new misplacement	Terminal repair quality and collateral damage (E10)
RoboRecover [2]	Recovery by stage and source/recovery policy	Same-prefix effect, stage, same-state recovery (E2/E6/E12)
BrickBench [1]	Validity, semantic quality, human judgment, cost	Separate legality, exact goal, human results, and cost (E1/E5/E8)

Prerequisites and budgets. Complete image rendering and blind readability review before S visual scoring. Verify the observation adapter, trajectory saving, actual-event logging, and both cloned continuations on deterministic smoke tests before paid trials. For this planned comparison, use a decision cap $\max(40, 6N + 20)$ for every track, with N target pieces; retain M's additional $4N + 12$ manipulation cap. Repeated inspection and malformed outputs consume decisions. S's feasibility witnesses remain reference costs, not optimality claims. Store input images, actual actions, accepted/rejected moves, termination reason, and service errors. Retry transient transport failures at most twice with the same request; do not retry semantically incorrect model answers into success.

Pair construction and exposure. At the first admissible fault opportunity, save the environment and visible conversation once. Continue a control copy without an edit and a fault copy with the prescribed actual-state edit. A separate control run with the same seed is not this pair. D's 72 control definitions are additional construction controls; each of its 168 fault jobs still needs its own cloned control continuation. Keep scheduled faults with no admissible prefix in the exposure denominator, and report their stopping reasons. Count every API call in both continuations. The base plan has 5,150 static attempts and 2,880 dynamic shared-prefix or normal jobs across ten models with one run per case (8,030 work units, not API calls). There are 2,040 fault jobs, each requiring two continuations when the prefix is reached. Ablations, the human-comparison panel, detection probes, cross-model recovery, strategy/visual tests, admission, and full-bank coverage are additional work. Overlapping identical runs are reused, not counted twice.

Reading the radar profiles. The aggregate profile in the main paper separates tracks and outcomes; Figure S8 separates error families within D. Plot one series per model only when all required estimates exist, without filling missing values or joining across missing axes. Both templates are bound to the same result fields as their tables and currently contain no model curves. Rates use source-macro aggregation; their denominators differ across tasks, so neither polygon area nor an average of axes is a score. Exposure is an opportunity for repair, not evidence that the error was recognized. Tables remain the source of counts and uncertainty.

Table S10. **E4: dynamic error breakdown (unmeasured).** Instantiate one panel per named model in Table S8. Each D family has 14 fault definitions and one run per case. M supports only missing, color, and extra-layer faults; it has 12 definitions per family. Report actual exposure, recovery, and source-clustered intervals separately.

Track / mechanism	Definitions	Exposed / assigned	Recovery	95% CI
D: Removed piece	14	—	—	—
D: Shifted piece	14	—	—	—
D: Cascaded removal	14	—	—	—
D: Extra layer	14	—	—	—
D: Early cover	14	—	—	—
D: Wrong color	14	—	—	—
D: Wrong size	14	—	—	—
D: Rotated piece	14	—	—	—
D: Swapped positions	14	—	—	—
D: Wrong part	14	—	—	—
D: Buried color	14	—	—	—
D: Compound fault	14	—	—	—
M: Missing	12	—	—	—
M: Wrong color	12	—	—	—
M: Extra layer	12	—	—	—

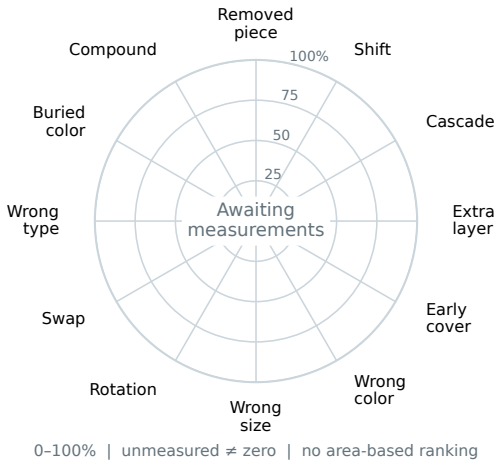

Figure S8. **Error-specific recovery profile template.** Each axis is conditional recovery for one D mechanism in Table S10. All axes use the same 0–100% scale. Missing measurements have no curve; an untriggered stratum is undefined, not zero. Report exposure counts and source-clustered intervals in the table.

Human comparison without written responses. Select twelve S control/fault pairs from different sources, twelve D pairs covering the twelve mechanisms, and one pair per M module, cycling evenly through its three errors. This yields 36 pairs or 72 conditions. Six independent participants each operate one condition from every pair, using buttons and graphical actions. Each condition receives three observations; nobody sees both members of the same pair. Randomize order and balance normal/fault assignments across participants. The exact schedule is saved before collection. Train on separate practice scenes. Record completion, exposure, actions, time, and an optional "cannot see / cannot operate" button. This is not three independent records when only one person submits an answer. The same panel is evaluated with all ten models once under the same views, action limits, and seed 41. Human estimates additionally cluster by participant.

Ablations and analysis units. Use GPT-6.1 Sol, Gemini 3.8 Flash, and Qwen3.5-27B for diagnostic experiments, selected before viewing outcomes. Apply no-manual only to M, no-previous-view only to dynamic tracks, and keep all other inputs and cases fixed. The privileged symbolic condition may expose actual part poses but must not silently provide intervention labels or recovery actions. Recompute both control and fault continuations when input changes alter the prefix. Report source-macro completion and recovery, as well as aggregate exposure numerator/denominator; do not average undefined stratum rates as zero. Use 2,000 bootstrap replicates clustered by source, retaining case variants together; paired differences resample the same sources. Sparse categories remain descriptive.

Do not infer a detection miss solely from an incomplete object. If anomaly judgments are measured, run a separately declared judgment panel at recorded observations without interrupting the primary assembly runs, and retain no-change controls. A state-query ablation and action trace inspection distinguish visible evidence, planning, and action selection, but their causal scope depends on what information changes. Estimate money from a small current-interface pilot for each selected model. Old car charges are archived observations, not current price quotes. The experi-

Table S11. **E5: independent human/model common panel (unmeasured).** Same 72 conditions for all agents. Six humans yield three independent observations per condition; models have one run per case. Report static completion, D/M normal completion (C), exposure (T), recovery (R), static false alarms (FA), and median active seconds. This panel is separate from primary-table denominators.

Agent	S	D C	D T	D R	M C	M T	M R	S FA	Time
Independent humans	—	—	—	—	—	—	—	—	—
GPT-6.1 Sol	—	—	—	—	—	—	—	—	—
Claude Sonnet 5.5	—	—	—	—	—	—	—	—	—
Gemini 3.8 Flash	—	—	—	—	—	—	—	—	—
Claude Haiku 5.5	—	—	—	—	—	—	—	—	—
Qwen3.5-9B	—	—	—	—	—	—	—	—	—
Qwen3.5-27B	—	—	—	—	—	—	—	—	—
InternVL3.5-8B	—	—	—	—	—	—	—	—	—
Qwen2.5-VL-7B	—	—	—	—	—	—	—	—	—
Qwen2.5-VL-32B	—	—	—	—	—	—	—	—	—
Gemma 3-12B	—	—	—	—	—	—	—	—	—

Table S12. **E6: difficulty and source breakdown (unmeasured).** Instantiate per track and named model. Part bands include the 16-piece S tasks; stage and temporary-removal rows apply to D/M. Dev/test comparisons use identical preregistered prompts, no tuning on test; M remains public development. Bands are not measured easy/hard labels.

Stratum	n	Exposed	Success/R
4–7 parts	—	—	—
8–10 parts	—	—	—
11–16 parts	—	—	—
Early / middle / late	—	—	—
No temporary removal	—	—	—
Temporary removal	—	—	—
Source dev / source test	—	—	—

Table S13. **E7: failure ledger (unmeasured).** Per named model and track; attempts may have multiple counters. Do not infer visual detection failure from an incomplete object.

Observed event	Count	Unit
Provider / transport failure	—	requests
Invalid action format	—	decisions
Rejected physical action	—	attempts
Fault not exposed	—	fault jobs
Wrong finish / budget end	—	episodes
Control pass, fault fail	—	exposed pairs
Wrong anomaly judgment	—	probe items

ment guide specifies caps and a cost formula; no paid calls are made in this revision.

I.1. Additional questions and reserved analyses

Paired intervention effect (E2). For each exposed pair, let $c, f \in \{0, 1\}$ denote control and fault completion. Save $n_{11}, n_{10}, n_{01}, n_{00}$, including cases where an edit helps rather than harms. The within-source difference is $\Delta = C_c - C_f$; average source differences and bootstrap sources jointly for its interval. Unexposed jobs remain in the exposure report and do not enter this conditional effect. Raw pooled counts and source-macro rates are both reported; the latter need not equal pooled ratios. Recovery conditional on control success is $n_{11}/(n_{11} + n_{10})$ within a source; retain its denominator.

Static correction quality (E9–E10). The error breakdown uses the actual 467-question S test assignments, including controls and migrated replacement states. Report M-static separately. Let G be the target multiset of part type, color, and pose; A_0, A_T the initial and terminal multisets, excluding locked context. Define $d(A, G) = |A \setminus G| + |G \setminus A|$. Defect reduction is $[d(A_0, G) - d(A_T, G)]/d(A_0, G)$ and may be negative. It is undefined for an initially correct control. Part F1 is $2|A_T \cap G|/(|A_T| + |G|)$; exact completion and physical legality remain the primary criteria. For initially correct placements $H = A_0 \cap G$, terminal collateral damage is $|H \setminus A_T|/|H|$, undefined when H is empty. Temporary removal followed by correct replacement causes no terminal damage. Count accepted state-changing edits on normal controls as unnecessary edits. These multiset metrics adapt SCANet's correction/damage distinction; they are not its original component-wise scores. Dynamic completion uses E1/E2, avoiding confusion between unfinished construction and a repair defect.

Detection independent of assembly (E11). Use certified fixed checkpoints from the twelve D and twelve M control/fault pairs in the human panel: 48 conditions with both changed and unchanged scenes. Every model sees the same before/after images, target, and public part/slot handles. Ask for change/no-change and a selection of affected handles; humans use buttons. Score change balanced accu-

Table S14. **E8: efficiency (unmeasured).** Instantiate separately by track. Count shared prefixes, both continuations, retries, input/output tokens, and hosted charges or measured GPU-hours. R_c: fault recovery conditional on exposed pairs whose control succeeds; n_c is its denominator. Report medians and dispersion for counts/time.

Model	R_c	n_c	Decisions	Inspects	Rejected	Seconds	USD
GPT-6.1 Sol	—	—	—	—	—	—	—
Claude Sonnet 5.5	—	—	—	—	—	—	—
Gemini 3.8 Flash	—	—	—	—	—	—	—
Claude Haiku 5.5	—	—	—	—	—	—	—
Qwen3.5-9B	—	—	—	—	—	—	—
Qwen3.5-27B	—	—	—	—	—	—	—
InternVL3.5-8B	—	—	—	—	—	—	—
Qwen2.5-VL-7B	—	—	—	—	—	—	—
Qwen2.5-VL-32B	—	—	—	—	—	—	—
Gemma 3-12B	—	—	—	—	—	—	—

Table S15. **E9: static error breakdown (unmeasured).** Counts are definitions in the current 467-question S test split, not the 3,292-question bank. Instantiate per named model, with one run per case and source-macro completion. Sparse categories are descriptive; repair-state replacements are reported separately.

S category	Definitions	Completed / trials	Success	95% CI
Compound	147	—	—	—
Normal control	23	—	—	—
Extra piece	2	—	—	—
Layer height	13	—	—	—
Missing	95	—	—	—
Offset	38	—	—	—
Replacement repair state	17	—	—	—
Rotation	35	—	—	—
Swapped positions	3	—	—	—
Wrong color	71	—	—	—
Wrong size	6	—	—	—
Wrong type	17	—	—	—

racy, anomaly recall, false alarms, and affected-slot F1. No-change is not a fault label, and removed pieces remain identifiable through previous-state slots. Three independent human ratings per condition also mark visibility and interface usability; ambiguous snapshots are reported and revised before a new version is evaluated. API validity is recorded separately. This probe is a separate task: it neither interrupts nor diagnoses an individual assembly trajectory with certainty. Its adapter and independent ratings remain to be built and collected.

Same-state cross-model repair (E12). Use the normal siblings of the twelve D and twelve M panel pairs, with no external edit during source collection. For each source model, save its first accepted attachment adding a placement outside the target multiset, then seek a legal recovery witness. This is an off-target state, not proof the original episode would fail; a temporary placement could still be corrected. If no such state arises or no witness can be found, retain the non-admission reason; do not substitute a later error from that run. Do not choose states for how well a recoverer performs. Each recoverer starts the identical state with the same target, remaining inventory, visible observations, and fresh context; source identity is hidden. Separate D/M panels and source-model rows reveal dependence on who created the failure. A diagonal restart differs from the original uninterrupted self-repair, which is reported separately. The state bank does not yet exist; admission counts and this selected-state scope must accompany every rate. This design follows cross-policy recovery analysis, without borrowing the physical claims of a robot benchmark.

Observation robustness (E13). Use the same 48-condition D/M panel for three diagnostic models. Change one input factor at a time: current image only, one fixed camera, or half/double image resolution. One-view removes alternate views from inspection as well as the initial prompt. Keep scene geometry and camera poses fixed for

Table S16. **E10: correction quality and collateral damage (unmeasured).** For S and M-static, report terminal defect reduction (DR), target-multiset part F1, exact completion, new damage to initially correct placements (CD), and unnecessary edits on normal controls (UE). Report S and M-static separately; dynamic recovery is reported separately in E1/E2. Temporary removals are allowed; CD measures terminal damage only.

Model	DR ↑	Part F1 ↑	Exact ↑	CD ↓	UE ↓
GPT-6.1 Sol	—	—	—	—	—
Claude Sonnet 5.5	—	—	—	—	—
Gemini 3.8 Flash	—	—	—	—	—
Claude Haiku 5.5	—	—	—	—	—
Qwen3.5-9B	—	—	—	—	—
Qwen3.5-27B	—	—	—	—	—
InternVL3.5-8B	—	—	—	—	—
Qwen2.5-VL-7B	—	—	—	—	—
Qwen2.5-VL-32B	—	—	—	—	—
Gemma 3-12B	—	—	—	—	—

Table S17. **E11: separate detection/localization probe (unmeasured).** Fixed certified snapshots from the 24 D/M control/fault pairs give 48 conditions. Queries never interrupt assembly. BA: balanced change-detection accuracy; TPR/FPR: anomaly recall/false alarms; Loc: affected piece/slot F1. Independent human readability is recorded with the same images.

Agent	BA	TPR	FPR	Loc	Readable
Independent humans	—	—	—	—	—
GPT-6.1 Sol	—	—	—	—	—
Claude Sonnet 5.5	—	—	—	—	—
Gemini 3.8 Flash	—	—	—	—	—
Claude Haiku 5.5	—	—	—	—	—
Qwen3.5-9B	—	—	—	—	—
Qwen3.5-27B	—	—	—	—	—
InternVL3.5-8B	—	—	—	—	—
Qwen2.5-VL-7B	—	—	—	—	—
Qwen2.5-VL-32B	—	—	—	—	—
Gemma 3-12B	—	—	—	—	—

Table S18. **E12: same-state cross-model recovery (unmeasured).** Rows identify the agent that created an off-target placement; columns identify the recoverer. Sol, Flash, and Qwen denote the three diagnostic models. Save the first off-target attachment in normal panel cases, certify recovery; restart all recoverers with identical visible context. Report admitted-state n and recovery separately for D/M. A diagonal restart is not the original uninterrupted self-repair.

State source	n	Sol	Flash	Qwen
Sol	—	—	—	—
Flash	—	—	—	—
Qwen 27B	—	—	—	—

resolution tests; verify provider preprocessing and record actual delivered image sizes and token use. E3 already covers manual, candidate-preview, feedback, and privileged-state conditions; reuse identical baseline runs. Recheck visibility under each condition and report results for all assigned items and the jointly readable subset. These tests change available evidence; they cannot automatically be described as changes in reasoning ability.

Planning and computation (E14). Compare the standard one-action closed loop to a single generated action script with no intermediate observation returned, an explicit visual checking instruction using the same public images/history, and twice the decision allowance. All scripts execute one brick at a time through the same legality checks; use deterministic part/pose references, never a hidden reference plan. Keep the manipulation cap fixed in the doubled condition. State which native reasoning effort is supported and record actual tokens, time, and charges; equal action limits do not imply equal provider computation. No private reasoning trace is scored. The script adapter and observation variants are planned, not implemented results.

Table S19. **E13: observation robustness (unmeasured).** Same 48-condition D/M panel; instantiate per diagnostic model and track. Base views are 900×660; half/double resolution changes only image sampling. One-view hides all alternate views, including inspection. Compare visibility admission before attributing any drop to reasoning.

Observation	Exposed / assigned	Recovery	Decisions	Input tokens
Full, 900 x 660	—	—	—	—
Current image only	—	—	—	—
One fixed view only	—	—	—	—
Half, 450 x 330	—	—	—	—
Double, 1800 x 1320	—	—	—	—

Table S20. **E14: planning strategy and budget (unmeasured).** Per diagnostic model and track on the same panel. Open-loop emits one script, still executed one brick at a time. The explicit-check variant uses the same images and public action history; it receives no error label. Double budget changes only the decision cap and retains the manipulation cap. Record native reasoning settings; no private reasoning trace is scored.

Strategy	Completion	Exposure	Recovery	Decisions	Seconds	USD
Closed-loop baseline	—	—	—	—	—	—
One-shot open-loop script	—	—	—	—	—	—
Explicit visual check	—	—	—	—	—	—
Double decision budget	—	—	—	—	—	—

J. Rendering, Attribution, and Physical Limits

Publication images. All brick scenes in the paper are rendered from original geometry at 3200×2400 pixels, with $2\times$ supersampling in each dimension before downsampling. Composite figure exports use 600 dpi. PDF and SVG retain vector labels, arrows, tables, and process boxes. The paper's larger renders do not replace the 900×660 images received by the current community interface. Orange boxes and manual highlights are presentation annotations and are disclosed in the relevant captions. They are not supplied as automatic fault detections.

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Physical scope. The environment checks rigid geometry, supported connections, and straight manipulation paths. Unsupported surrounding parts are retained as conservative collision bounds rather than deleted, which can reject some motions that a more precise mesh would allow. The preassembled context is fixed. The executor does not model deformation, insertion force, frictional stability, accidental slipping, articulated mechanics, or a robot gripper. Consequently, "legal" and "recoverable" refer to this geometric contract. Neither the protocol study nor its renders establish real-robot performance.

Table S21. **Complete static configuration inventory (1/2).** P: target parts; Q: current questions. Identifiers refer to the accompanying full model catalogue.

Index	Model/configuration	P	Q	Index	Model/configuration	P	Q
S001	10001	11	72	S041	v23-10233	11	23
S002	10001-deep	16	48	S042	v23-10241	11	18
S003	10014	11	33	S043	v23-10248	11	25
S004	10036	11	73	S044	v23-10258	11	18
S005	10036-alt1	11	28	S045	v23-10261	11	17
S006	10128	11	24	S046	v23-10262	11	22
S007	10156	11	68	S047	v23-10267	11	18
S008	10156-alt1	11	20	S048	v23-10271	11	21
S009	10159	11	23	S049	v23-1472	11	18
S010	10213	11	62	S050	v23-1484	11	16
S011	10214	11	21	S051	v23-1489	11	23
S012	10220	11	28	S052	v23-1490	11	16
S013	31009	11	18	S053	v23-1490-alt1	11	27
S014	31027	11	22	S054	v23-1572	11	22
S015	31027-alt1	11	26	S055	v23-1572-alt1	11	23
S016	31027-kart	11	22	S056	v23-1632	11	19
S017	31028	11	6	S057	v23-1633	11	23
S018	31028-alt1	11	25	S058	v23-1682	11	16
S019	31028-sailboat	11	25	S059	v23-1775	11	16
S020	5867	11	22	S060	v23-1775-alt1	10	21
S021	omr-10242	11	16	S061	v23-1792	11	22
S022	omr-10265	11	9	S062	v23-1808	11	22
S023	omr-10269	11	3	S063	v23-1808-alt1	11	21
S024	omr-21309	11	19	S064	v23-1854	11	18
S025	omr-31031	11	20	S065	v23-1854-alt1	11	25
S026	omr-31088	11	6	S066	v23-20015	11	21
S027	omr-42004	11	18	S067	v23-20015-alt1	11	25
S028	omr-42020	11	18	S068	v23-21004	11	14
S029	omr-42020-alt1	11	17	S069	v23-21005	11	17
S030	omr-42054	11	18	S070	v23-21006	11	22
S031	omr-42061	11	18	S071	v23-21007	11	18
S032	omr-42066	11	7	S072	v23-21009	11	14
S033	omr-42102	11	17	S073	v23-21010	11	22
S034	omr-42102-alt1	11	9	S074	v23-21011	11	21
S035	v23-10037	11	15	S075	v23-21012	11	22
S036	v23-10124	11	22	S076	v23-21014	11	22
S037	v23-10158	11	29	S077	v23-21016	11	18
S038	v23-10184	11	17	S078	v23-21017	11	22
S039	v23-10194	11	17	S079	v23-21018	11	22
S040	v23-10226	11	24	S080	v23-21020	11	17

Table S22. **Complete static configuration inventory (2/2).** P: target parts; Q: current questions. Identifiers refer to the accompanying full model catalogue.

Index	Model/configuration	P	Q	Index	Model/configuration	P	Q
S081	v23-21021	11	22	S121	v23-31040	11	5
S082	v23-21022	11	22	S122	v23-31040-alt1	11	26
S083	v23-21024	11	22	S123	v23-31045	11	17
S084	v23-21029	11	22	S124	v23-31048	11	22
S085	v23-21030	11	22	S125	v23-31049	11	7
S086	v23-21037	11	22	S126	v23-31063	11	23
S087	v23-21045	11	22	S127	v23-31066	11	23
S088	v23-21302	11	17	S128	v23-31070	11	8
S089	v23-21303	11	21	S129	v23-31072	11	22
S090	v23-21318	11	22	S130	v23-31072-alt1	11	22
S091	v23-21319	11	23	S131	v23-31084	11	22
S092	v23-21321	11	17	S132	v23-31092	11	11
S093	v23-21325	11	21	S133	v23-31092-alt1	11	22
S094	v23-30022	7	1	S134	v23-3178	11	18
S095	v23-30023	7	20	S135	v23-3178-alt1	11	22
S096	v23-30025	11	11	S136	v23-3179	11	5
S097	v23-30025-alt1	11	20	S137	v23-3179-alt1	11	26
S098	v23-30026	11	21	S138	v23-3181	11	18
S099	v23-30026-alt1	11	20	S139	v23-3221	11	23
S100	v23-30027	11	8	S140	v23-3223	11	22
S101	v23-30027-alt1	11	6	S141	v23-3223-alt1	8	18
S102	v23-30029	11	22	S142	v23-3223-deep	16	27
S103	v23-30029-alt1	11	22	S143	v23-4018	9	21
S104	v23-30181	11	22	S144	v23-6678	11	22
S105	v23-30181-alt1	11	7	S145	v23-6678-deep	16	23
S106	v23-30285	11	3	S146	v23-6679	11	16
S107	v23-30311	11	22	S147	v23-6682	11	22
S108	v23-30311-alt1	11	24	S148	v23-6682-alt1	11	21
S109	v23-30312	11	21	S149	v23-6683	11	22
S110	v23-30472	11	15	S150	v23-6683-alt1	8	18
S111	v23-30472-alt1	11	24	S151	v23-6689	11	23
S112	v23-30542	11	12	S152	v23-6689-alt1	11	25
S113	v23-30542-alt1	11	20	S153	v23-6690	11	22
S114	v23-30571	11	22	S154	v23-6690-alt1	11	8
S115	v23-30571-alt1	11	25	S155	v23-6698	11	10
S116	v23-30571-deep	16	16	S156	v23-6698-alt1	11	24
S117	v23-31025	11	17	S157	v23-6699	11	17
S118	v23-31037	11	18	S158	v23-6699-alt1	11	18
S119	v23-31038	11	17	S159	v23-6745	11	22
S120	v23-31039	11	23	S160	v23-6753	11	23

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